BANK MANAGEMENT SYSTEM

TEST PLAN DOCUMENT

Manual test cases covering unit (UT), integration (IT), and system (ST) testing for all eight functional requirements. Actual Result and Test Result are left blank for completion after manual execution against the running system.

1. FR-1: Open New Account

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0101	Account Management	Verify that the system can open account with valid details.	Customer is on the account-opening page.	1. Enter valid name, address, phone and ID proof 2. enter ₹500 3. submit.	Valid KYC; deposit ₹500	Account is created successfully and the system proceeds to account-number generation.		
UT_0102	Account Management	Verify that the system can reject initial deposit below ₹500.	Customer is on the account-opening page.	1. Enter valid KYC 2. enter ₹499 3. submit.	Valid KYC; deposit ₹499	Request is rejected; no account is created.		
UT_0103	Account Management	Verify that the system can accept initial deposit above ₹500.	Customer is on the account-opening page.	1. Enter valid KYC 2. enter ₹1,000 3. submit.	Valid KYC; deposit ₹1,000	Account-opening request is accepted.		
UT_0104	Account Management	Verify that the system can reject blank customer name.	Customer is on the account-opening page.	1. Leave name blank 2. complete other fields 3. submit.	Name: blank; deposit ₹500	Validation message is shown and no account is created.		
UT_0105	Account Management	Verify that the system can reject blank address.	Customer is on the account-opening page.	1. Leave address blank 2. complete other fields 3. submit.	Address: blank; deposit ₹500	Address validation is shown and no account is created.		
UT_0106	Account Management	Verify that the system can reject invalid phone number.	Customer is on the account-opening page.	1. Enter an invalid phone number and submit.	Phone: 12AB; deposit ₹500	Phone-number validation is shown; account is not created.		
IT_0107	Account Management	Verify that the system can reject missing id proof.	Customer is on the account-opening page.	1. Do not provide ID proof 2. submit the form.	ID proof: blank; deposit ₹500	ID-proof validation is shown; account is not created.		
IT_0108	Account Management	Verify that the system can reject zero initial deposit.	Customer is on the account-opening page.	1. Enter valid KYC and ₹0 2. submit.	Deposit ₹0	Request is rejected because the deposit is below ₹500.		
ST_0109	Account Management	Verify that the system can reject negative initial deposit.	Customer is on the account-opening page.	1. Enter valid KYC and a negative deposit 2. submit.	Deposit: -₹100	Request is rejected and no account is created.		
ST_0110	Account Management	Verify that the system can verify account details after creation.	Valid KYC details and minimum deposit are available.	1. Open an account 2. search for the newly created customer record.	Valid KYC; deposit ₹500	Saved customer details can be retrieved correctly.		

2. FR-2: Generate Unique 10-Digit Account Number

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0201	Account Number Generation	Verify that the system can generate a 10-digit account number.	A valid account-opening request is accepted.	1. Complete account opening and inspect the assigned number.	New valid account	A 10-digit account number is generated.		
UT_0202	Account Number Generation	Verify that the system can verify account number contains digits only.	A valid account-opening request is accepted.	1. Create an account and inspect each character of its number.	New valid account	Account number contains exactly 10 numeric digits.		
UT_0203	Account Number Generation	Verify that the system can verify account number uniqueness.	At least one account already exists.	1. Create another account and compare its number with existing numbers.	Two valid account requests	New account number differs from every existing account number.		
UT_0204	Account Number Generation	Verify that the system can regenerate on duplicate candidate.	Test setup can simulate a duplicate generated candidate.	1. Cause generator to produce an existing number 2. complete generation.	Existing number as first candidate	System rejects the duplicate candidate and generates another unique number.		
UT_0205	Account Number Generation	Verify that the system can check number stored with account.	A valid account-opening request is accepted.	1. Create an account 2. retrieve its record from account list/database.	New valid account	The generated number is saved against the correct customer.		
UT_0206	Account Number Generation	Verify that the system can check number remains stable.	An account already exists.	1. View the same account in two separate requests.	Existing account number	The account number remains unchanged between views.		
IT_0207	Account Number Generation	Verify that the system can generate numbers for multiple accounts.	System is operational.	1. Create five valid accounts and record their numbers.	Five valid account requests	All generated numbers are 10 digits and mutually unique.		
IT_0208	Account Number Generation	Verify that the system can do not generate number for rejected application.	Account-opening form is available.	1. Submit an application with deposit ₹400 2. inspect account records.	Invalid initial deposit ₹400	No account number is generated for the rejected application.		
ST_0209	Account Number Generation	Verify that the system can handle generator collision safely.	Generator collision can be simulated in test environment.	1. Create an account while the first generated candidate collides with an existing number.	Collision followed by unique candidate	System retries safely and assigns a unique number.		
ST_0210	Account Number Generation	Verify that the system can prevent duplicate number at data-store level.	An existing account number is stored.	1. Attempt to save a second account using that same number.	Duplicate 10-digit account number	Duplicate is rejected; existing account data remains unchanged.		

3. FR-3: View, Edit and Deactivate Customer Account

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0301	Account Management	Verify that the system can view existing account.	Staff/admin is authenticated and account exists.	1. Search by account number and open the account.	Existing active account	Correct customer and account details are displayed.		
UT_0302	Account Management	Verify that the system can view non-existent account.	Staff/admin is authenticated.	1. Search using an account number that does not exist.	Account number: 9999999999	A not-found message is displayed; no unrelated record appears.		
UT_0303	Account Management	Verify that the system can edit customer address.	An existing account is available.	1. Open account 2. edit address 3. save 4. reopen record.	New address: Bengaluru	Updated address is saved and displayed.		
UT_0304	Account Management	Verify that the system can edit customer phone number.	An existing account is available.	1. Edit phone number with a valid value and save.	Valid new 10-digit phone	Phone number is updated successfully.		
UT_0305	Account Management	Verify that the system can reject invalid phone during edit.	An existing account is available.	1. Edit phone number to an invalid value and save.	Phone: 12AB	Validation is shown and old phone number is retained.		
UT_0306	Account Management	Verify that the system can cancel account edit.	An existing account is available.	1. Change a field 2. select Cancel without saving.	Temporary address change	Original account details remain unchanged.		
IT_0307	Account Management	Verify that the system can deactivate account with confirmation.	An active account exists and staff/admin is authenticated.	1. Open account 2. select Deactivate 3. confirm.	Existing active account	Account status changes to inactive/deactivated.		
IT_0308	Account Management	Verify that the system can cancel deactivation.	An active account exists.	1. Select Deactivate and cancel confirmation.	Existing active account	Account remains active.		
ST_0309	Account Management	Verify that the system can prevent unauthorized account edit.	User without staff/admin permissions is logged in.	1. Attempt to open edit action and save a change.	Unauthorized user; existing account	Access is denied and account details remain unchanged.		
ST_0310	Account Management	Verify that the system can verify deactivated account status.	An account has been deactivated.	1. Search for the account and inspect its status.	Deactivated account	Account is identifiable as inactive and is not shown as active.		

4. FR-4: Validate Withdrawal and Minimum Balance

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0401	Withdrawal	Verify that the system can reject withdrawal below minimum balance.	Active account balance is ₹1,500.	1. Request withdrawal of ₹600.	Balance ₹1,500; withdrawal ₹600	Withdrawal is rejected because remaining balance would be ₹900; balance is unchanged.		
UT_0402	Withdrawal	Verify that the system can allow withdrawal retaining minimum balance.	Active account balance is ₹2,000.	1. Request withdrawal of ₹1,000.	Balance ₹2,000; withdrawal ₹1,000	Withdrawal succeeds and balance becomes ₹1,000.		
UT_0403	Withdrawal	Verify that the system can allow small valid withdrawal.	Active account balance is ₹2,000.	1. Request withdrawal of ₹500.	Balance ₹2,000; withdrawal ₹500	Withdrawal succeeds and balance becomes ₹1,500.		
UT_0404	Withdrawal	Verify that the system can reject withdrawal equal to full balance.	Active account balance is ₹5,000.	1. Request withdrawal of ₹5,000.	Balance ₹5,000; withdrawal ₹5,000	Withdrawal is rejected because the minimum balance would not be maintained.		
UT_0405	Withdrawal	Verify that the system can reject withdrawal exceeding balance.	Active account balance is ₹2,000.	1. Request withdrawal of ₹2,500.	Balance ₹2,000; withdrawal ₹2,500	Withdrawal is rejected and balance remains ₹2,000.		
UT_0406	Withdrawal	Verify that the system can reject zero withdrawal.	Active account balance is ₹2,000.	1. Enter ₹0 and submit withdrawal.	Withdrawal ₹0	Validation is shown; no transaction is recorded.		
IT_0407	Withdrawal	Verify that the system can reject negative withdrawal.	Active account balance is ₹2,000.	1. Enter -₹100 and submit withdrawal.	Withdrawal -₹100	Validation is shown; no balance change occurs.		
IT_0408	Withdrawal	Verify that the system can reject withdrawal from inactive account.	Account is deactivated.	1. Attempt to withdraw ₹100.	Inactive account; ₹100	Withdrawal is denied and balance remains unchanged.		
ST_0409	Withdrawal	Verify that the system can check balance after successful withdrawal.	Active account balance is ₹3,000.	1. Withdraw ₹1,000 2. view balance and history.	Balance ₹3,000; withdrawal ₹1,000	Balance is ₹2,000 and the withdrawal is recorded.		
ST_0410	Withdrawal	Verify that the system can prevent concurrent withdrawals causing invalid balance.	Account balance is ₹1,500 and two sessions can submit transactions.	1. Submit two ₹500 withdrawals concurrently.	Balance ₹1,500; two withdrawals of ₹500	Transactions are processed safely; balance never falls below ₹1,000.		

5. FR-5: Generate Transaction ID and Receipt

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0501	Deposit / Transaction Management	Verify that the system can deposit generates transaction id and receipt.	Active account has ₹5,000.	1. Deposit ₹2,000 and open receipt.	Deposit ₹2,000	Deposit succeeds; unique transaction ID and receipt show balance ₹7,000.		
UT_0502	Deposit / Transaction Management	Verify that the system can withdrawal generates transaction id and receipt.	Active account has ₹2,000.	1. Withdraw ₹500 and open receipt.	Withdrawal ₹500	Withdrawal succeeds; unique transaction ID and receipt show balance ₹1,500.		

UT_0503	Deposit / Transaction Management	Verify that the system can receipt includes transaction date and time.	A valid transaction can be completed.	1. Complete a deposit and inspect receipt fields.	Deposit ₹1,000	Receipt contains transaction date/time.		
UT_0504	Deposit / Transaction Management	Verify that the system can receipt includes transaction type.	A valid deposit or withdrawal can be completed.	1. Complete a deposit and inspect receipt.	Deposit ₹500	Receipt correctly identifies the transaction as Deposit.		
UT_0505	Deposit / Transaction Management	Verify that the system can receipt includes amount.	A valid transaction can be completed.	1. Complete a transaction and compare receipt amount.	Deposit ₹1,250	Receipt displays the exact transaction amount.		
UT_0506	Deposit / Transaction Management	Verify that the system can receipt includes post-transaction balance.	Account balance is ₹2,000.	1. Deposit ₹500 and inspect receipt.	Deposit ₹500	Receipt shows post-transaction balance ₹2,500.		
IT_0507	Deposit / Transaction Management	Verify that the system can transaction ids are unique.	At least one prior transaction exists.	1. Perform two separate successful transactions and compare IDs.	Two deposits of ₹500	Each successful transaction has a different transaction ID.		
IT_0508	Deposit / Transaction Management	Verify that the system can failed transaction does not produce success receipt.	Withdrawal would violate minimum balance.	1. Attempt an invalid withdrawal and inspect receipt/history.	Balance ₹1,200; withdrawal ₹500	No successful receipt is generated for the rejected transaction.		
ST_0509	Deposit / Transaction Management	Verify that the system can receipt corresponds to correct account.	Two active accounts exist.	1. Perform a deposit on one account and inspect receipt.	Deposit to account A	Receipt identifies the correct account/transaction without mixing account data.		
ST_0510	Deposit / Transaction Management	Verify that the system can receipt remains retrievable from history.	A transaction has completed successfully.	1. Open transaction history and select the transaction receipt.	Previously completed deposit	The correct receipt can be retrieved.		

6. FR-6: Lock Account After 3 Consecutive Failed Login Attempts

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0601	User Authentication / Login	Verify that the system can lock after three consecutive failures.	Account exists, active and unlocked.	1. Submit incorrect password three times consecutively.	Valid account ID; wrong password ×3	Account is locked for 15 minutes.		
UT_0602	User Authentication / Login	Verify that the system can allow login before three failures when correct credentials used.	Account exists and is unlocked.	1. Enter wrong password once, then correct password.	Wrong password once; correct password	Successful login resets the consecutive failure sequence.		
UT_0603	User Authentication / Login	Verify that the system can do not lock after two failures.	Account exists and is unlocked.	1. Submit incorrect password twice 2. check account status.	Wrong password ×2	Account remains unlocked; failed attempts are tracked.		
UT_0604	User Authentication / Login	Verify that the system can block login during lock period.	Account has just been locked.	1. Attempt login with correct credentials immediately.	Correct credentials during lock	Login is denied until the 15-minute lock expires.		
UT_0605	User Authentication / Login	Verify that the system can allow login after 15-minute lock expires.	Account was locked more than 15 minutes ago.	1. Enter correct credentials after lock expiry.	Correct credentials after 15 minutes	Login is allowed if credentials are valid.		
UT_0606	User Authentication / Login	Verify that the system can reset failure count after successful login.	Account has one failed attempt.	1. Log in successfully 2. then submit two failed attempts.	One failure, success, then two failures	The new failures are counted from zero; account is not locked at two.		
IT_0607	User Authentication / Login	Verify that the system can track failures for correct account.	Two accounts exist.	1. Enter account A with wrong password three times.	Account A and wrong password ×3	Only account A is locked; account B remains unaffected.		
IT_0608	User Authentication / Login	Verify that the system can reject unknown account id safely.	Login page is available.	1. Submit a non-existent account ID with a password three times.	Unknown account ID	Login fails safely without revealing sensitive account information.		
ST_0609	User Authentication / Login	Verify that the system can display clear lock message.	Account is unlocked at start.	1. Cause three consecutive failed login attempts.	Wrong password ×3	System displays a message that the account is locked for 15 minutes.		
ST_0610	User Authentication / Login	Verify that the system can successful login with valid credentials.	Account exists, active and unlocked.	1. Enter valid account ID and password 2. submit.	Correct account ID and password	User is authenticated and redirected to the appropriate dashboard.		

7. FR-7: Log Every Deposit and Withdrawal

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0701	Transaction Management / Transaction Log	Verify that the system can log successful deposit.	Active account exists.	1. Deposit ₹1,000 2. open transaction log.	Deposit ₹1,000	Log includes timestamp, Deposit type, amount and post-transaction balance.		
UT_0702	Transaction Management / Transaction Log	Verify that the system can log successful withdrawal.	Active account balance is ₹2,000.	1. Withdraw ₹500 2. open transaction log.	Withdrawal ₹500	Log includes timestamp, Withdrawal type, amount and balance ₹1,500.		
UT_0703	Transaction Management / Transaction Log	Verify that the system can log contains accurate timestamp.	System clock is configured.	1. Complete a transaction and inspect its log entry.	Deposit ₹100	Log timestamp corresponds to transaction time.		
UT_0704	Transaction Management / Transaction Log	Verify that the system can log correct transaction type.	Active account exists.	1. Perform one deposit and one withdrawal 2. inspect entries.	Deposit ₹200; withdrawal ₹100	Each entry is labelled with the correct transaction type.		
UT_0705	Transaction Management / Transaction Log	Verify that the system can log exact amount.	Active account exists.	1. Complete a transaction with a non-round amount.	Deposit ₹1,234	Log records the exact amount ₹1,234.		

UT_0706	Transaction Management / Transaction Log	Verify that the system can log correct post-transaction balance.	Account balance is ₹3,000.	1. Deposit ₹700 2. inspect transaction log.	Deposit ₹700	Log records post-transaction balance ₹3,700.		
IT_0707	Transaction Management / Transaction Log	Verify that the system can do not log rejected withdrawal as completed.	Balance is ₹1,200.	1. Attempt withdrawal ₹500 and inspect log.	Withdrawal ₹500	No completed withdrawal entry is recorded; balance remains unchanged.		
IT_0708	Transaction Management / Transaction Log	Verify that the system can maintain separate entries for multiple transactions.	Active account exists.	1. Perform two deposits and one withdrawal 2. view history.	₹500 deposit, ₹300 deposit, ₹200 withdrawal	Three distinct and correctly ordered entries appear.		
ST_0709	Transaction Management / Transaction Log	Verify that the system can keep transaction log associated with correct account.	Two active accounts exist.	1. Transact on account A and inspect logs for A and B.	Deposit to account A	Transaction appears only under the correct account.		
ST_0710	Transaction Management / Transaction Log	Verify that the system can verify log persists after reopening.	At least one transaction is logged.	1. Close/reopen the transaction history or log in again 2. inspect entry.	Previously completed transaction	Transaction log entry remains available and unchanged.		

8. FR-8: Generate Admin Report

Test Case ID	Name of Module	Test Case Description	Pre-Conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_0801	Admin Reports	Verify that the system can generate report for valid date range.	Admin is logged in and records exist.	1. Select start/end dates and generate report.	01-08-2026 to 31-08-2026	Report displays total deposits, total withdrawals and active account count.		
UT_0802	Admin Reports	Verify that the system can reject start date after end date.	Admin is logged in.	1. Set start date later than end date and generate.	Start 31-08-2026; end 01-08-2026	Report is not generated and date-range validation is displayed.		
UT_0803	Admin Reports	Verify that the system can generate report for date with no activity.	Admin is logged in.	1. Select a valid range containing no transactions and generate.	01-01-2020 to 31-01-2020	System displays a no-records message or a valid zero-activity report.		
UT_0804	Admin Reports	Verify that the system can verify total deposits.	Deposit records exist in selected range.	1. Generate report and compare total with source transactions.	Deposits ₹500 and ₹1,000	Reported deposit total equals ₹1,500.		
UT_0805	Admin Reports	Verify that the system can verify total withdrawals.	Withdrawal records exist in selected range.	1. Generate report and compare total with source transactions.	Withdrawals ₹200 and ₹300	Reported withdrawal total equals ₹500.		
UT_0806	Admin Reports	Verify that the system can verify active account count.	Active and inactive accounts exist.	1. Generate report and compare count against account statuses.	3 active and 2 inactive accounts	Report counts only the 3 active accounts.		
IT_0807	Admin Reports	Verify that the system can verify report date filtering.	Transactions exist both inside and outside selected range.	1. Generate report for a narrow date range.	Range with known transactions	Only records within the selected date range contribute to totals.		
IT_0808	Admin Reports	Verify that the system can generate report within 10 seconds.	Admin is logged in and report data exists.	1. Generate report and measure elapsed time.	Representative report dataset	Report is generated within 10 seconds.		
ST_0809	Admin Reports	Verify that the system can restrict report generation to admin.	A user without admin permissions is logged in.	1. Attempt to access and generate an admin report.	Non-admin user	Access is denied and report data is not exposed.		
ST_0810	Admin Reports	Verify that the system can verify report values against transaction log.	Transaction log contains known deposits and withdrawals.	1. Generate report for matching date range 2. compare totals.	Known transaction dataset	Report totals match the underlying transaction records.		